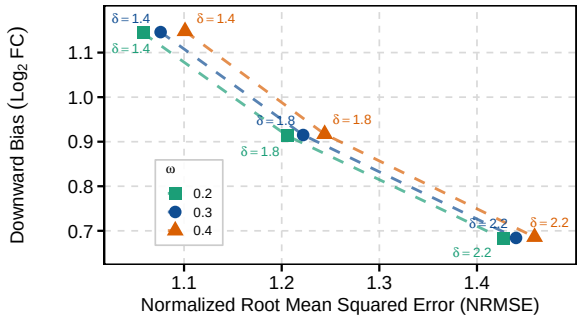
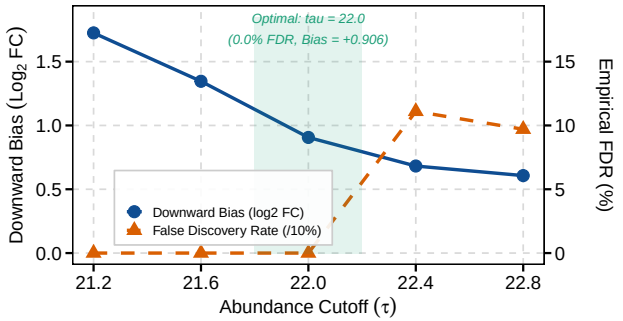
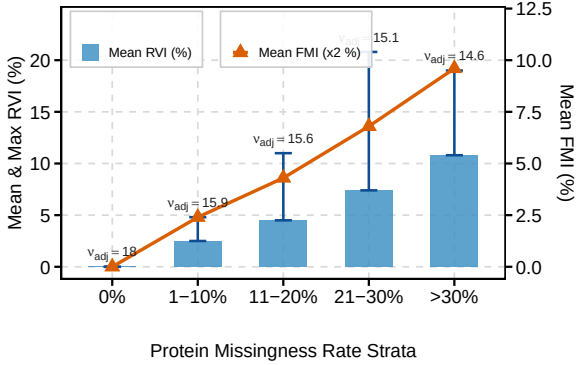

(a) Perseus Hyperparameter Tradeoff
Higher delta reduces bias; inflates NRMSE
Downward Bias (Log2 FC)
Normalized Root Mean Squared Error (NRMSE)
ω
0.2
0.3
0.4
δ = 1.4
δ = 1.8
δ = 2.2


(b) Hybrid Cutoff Optimization
tau = 22.0 balances MAR donors & MNAR tail
Optimal: tau = 22.0
(0.0% FDR, Bias = +0.906)
Downward Bias (log2 FC)
False Discovery Rate (/10%)
Downward Bias (Log2 FC)
Empirical FDR (%)
Abundance Cutoff (τ)

(c) Cross−Method Concordance
Single imputation methods cluster (J > 0.75)
Jaccard Index
Oracle
MinDet
Perseus
QRILC
kNN
Hybrid
Rubin_MI
proDA


(d) Rubin MI Information Loss & RVI
RVI to 20.8%, df shrinks 18.0 to 14.6
Mean RVI (%)
Mean FMI (x2 %)
Mean & Max RVI (%)
Mean FMI (%)
Protein Missingness Rate Strata
0%
1−10%
11−20%
21−30%
>30%